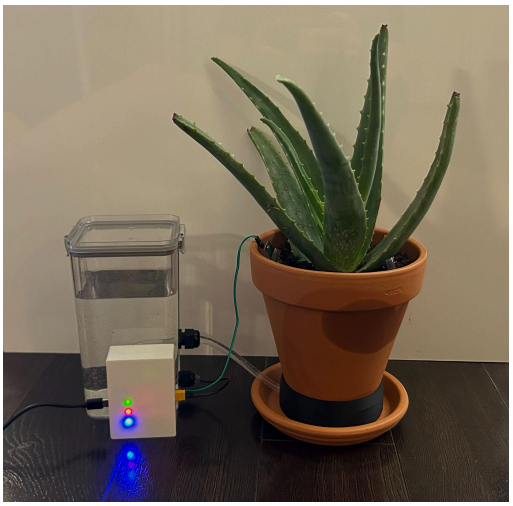
Aloe Vera

January 22/2026 - March

Kendra Smart



GitHub:
https://github.com/kesmart/AloeVera

OnShape:
https://cad.onshape.com/documents/b15bd51a204e2427939fe73d/w/d66d37e671b4fbe30a57183f/e/3a2b51e5b67cc1b45a7122af?renderMode=0&uiState=69b2faf8191458fa1230e08f

Description: I would like to make a pump-irrigation system for my aloe vera plant (and also move it to a larger pot) to water it on a timer or from my phone. I would also like to monitor the soil's moisture level so I can read when it is time to water it again.

Step 1 – Assemble irrigation system 2
Step 2 – Testing pump 3
Step 3 – Code & Web Server 4
Step 4 – Assemble full system 5
BOM 6
Brain Dump 7

Step 1 – Assemble irrigation system

Assembling the irrigation system in my pot – routing the tubes and testing water flow through it manually with a syringe. Need a 1 inlet, 3 outlet flow splitter and a base to route the tubing out through.

<u>Assembly:</u>

- Base piece fits together with the pot very nicely
- The turn radius for the input tube through the bottom section works but could be a bit bigger on a next iteration
  - ☐ Increase turn radius
- When I epoxied the tubes into the flow splitter, I accidently epoxied one of the paths closed so water doesn't go through that tube – oops
  - ☑ ~~Use less epoxy inside next time, add more on outside if necessary~~
- I changed the input tube size to match my pump, this needs to be changed for the base piece
  - ☐ Re-size base piece for larger input tube

<u>Function:</u>

- No leakage in the 3D printed flow splitter

Assembly of tubes in pot.

Step 2 – Testing pump

Testing the little pumps I bought

Wiring Diagram:
https://app.cirkitdesigner.com/project/1fcb4530-1d4d-4bc4-a394-ff442a23c446

Assembly:

- Breadboarding went well → follow the wiring diagram
    - I skipped the sensor and red/green LEDs while initially testing the pump, then tested them separately.

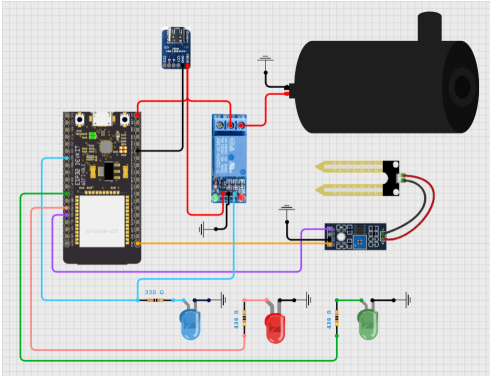
Function:

- When running the pump at 3.3V power, water only came out of 2/3 tubes. When I tested with the syringe I could recreate the problem but could also make the water come out of all 3 tubes with more pressure. It does work when powered at 5V though.
    - ☑ ~~Power pump with 5V~~
- Also tested the pump with the tubes upright to simulate the actual amount of head gain when the system is actually set up and it worked great.

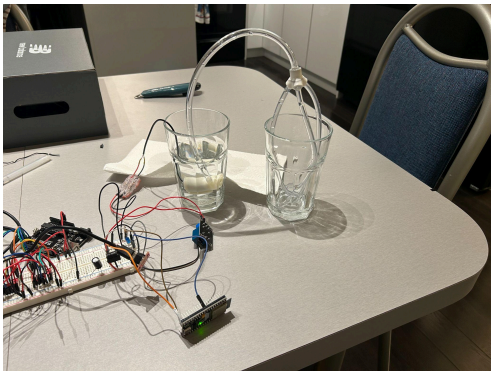
Conclusion:

- Works nicely

☐ Make perf board of the electronics.



Wiring diagram.



Testing pump.

Step 3 – Code & Web Server

https://randomnerdtutorials.com/esp32-web-server-arduino-ide/#:~:text=1)%20Plug%20your%20ESP32%20board,the%20%E2%80%9CDone%20uploading%E2%80%9D%20message.

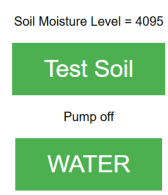
- Good starting point/ template for setting up and how to do a simple html code.
- Change what the buttons do and add moisture reading.

Notes:

- The web server has a delay when used on your phone. No delay on the laptop.
- Pauses and the button changes to "watering…" while watering. Needs to reload the page afterwards to change the button back

Future ideas:

- [ ] Use green/red lights to indicate moisture level.
    - Need measured values of dry, moist and wet soil as references.
    - Then choose the reference moisture where I want to indicate the soil is too dry.



Step 4 – Assemble full system

(1) Assemble the pump system into the pot (transfer Aloe Vera). (2) Get a water bottle and get the pump system into it. (3) Simplify electronics.

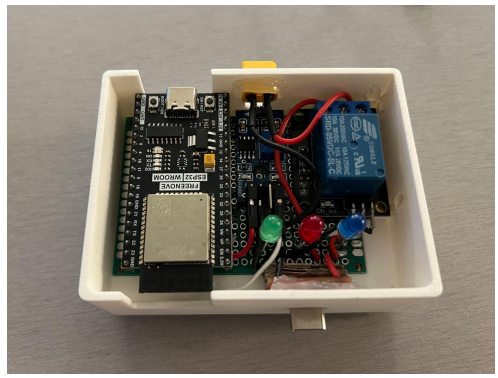
- Use hot glue to pot the wire and output of the pump for proof of concept then we can change it to epoxy.

<u>Perfboard Notes:</u>

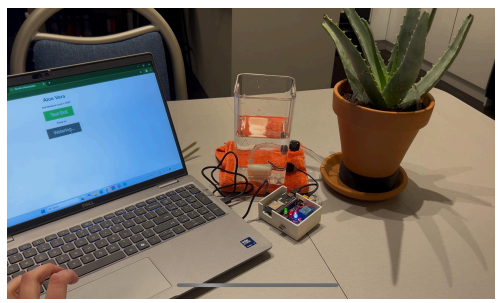
- The only problem is that the red, green, and blue LEDs are not lighting up.
  - Forgot to set the red and green pin modes as an output. They now light up. Voltage across lit up LEDs is 1.9V.
  - Blue LEDs have a higher forward voltage, V_f, than red and green and thus require a lower resistance.
    - This did not fix the issue.
  - I was using an IR Emitter, not a regular LED. They look so much alike! My bad 🙁
- Should maybe give more room in the case for the heat set inserts but I am not even using them so it's chill.

<u>Electronics case</u>

- Looks good, snap fit lid
- Never gonna be able to get bolts in the USB-C breakout board. I added pins and glued them instead.



Electronics box.



Successfully watering.

<u>Fully assembled system</u>

- The tank leaks from the wire reliefs, particularly the hose one
  - ☑ ~~Epoxy the wire reliefs/ penetrators~~
- Moisture sensor wires don't reach the box on the ground
  - ☐ Longer wires for the moisture sensor
- Tube ends are a bit weird; also one edge of the terracotta pot is wet and the other side is dry. I think all the water was sprayed to that side.
  - ☑ ~~Cut the sprinkler ends shorter.~~
  - ☐ Evenly orient all the sprinklers to water the entire plant.

BOM

Price includes the plant and potting supplies. A lot of these parts are things you might have laying around. Some of the more specific components from Amazon (relay, soil moisture sensor, ESP-32 Wroom, USB-C breakout, etc) can be swapped with similar alternatives that you have to save costs

Item	Description	Cost (CAD)
Aloe Vera	Aloe vera plant	6.98
Terra cotta pot	Plant pot.	6.48
Moisture control potting mix	Used ~1/8 of a $8.98 bag.	1.12
Mini Water Pump	Submersible water pump. Comes with tubing.	4.50 each
1/4" OD tubing	Used 3ft of this smaller tubing for the sprinklers. $10.20 for 20 ft of tube.	1.53
3D printed parts	Pot base, flow splitter, and electronics case are all 3D printed out of PLA. Total 108.21g of filament used. 1kg roll costs ~$19.99.	2.16
ESP-32 Wroom	2 for $26.95.	13.48
USB-C breakout board	Cheaper on Amazon. 10 pack for $14.29.	1.43
Relay module	$15.50 for a pack of 5.	3.10
Soil moisture sensor unit	$13.99 for a pack of 5.	2.80
3x LEDs	~$0.23 on DigiKey. You need a red, green, and blue.	0.69
2x 430 Ω resistors	Through hole resistors are all ~$0.15 on DigiKey	0.30
1x 330 Ω resistor		0.15
Perf board	About 18 x 20 holes. Not sure where I got it.	2.00
Female pin header strips	The link is to a whole set. This only uses a few	0.20
1x XT30 connector	12 pack for $13.49	1.12
Wire, heat shrink, solder		0.50
Total (including plant and potting supplies)		48.54
Total (not including plant and potting supplies)		33.96

Brain Dump

Soil Moisture Sensor

- Using analog input (tested using arduino uno 0-1023):
    - Prints 671 in very dry soil
    - Prints 320 when watered
        - Settled to 270 after 30 mins, then 280 after 2 hours
        - 450 after 24 hours

Water Pump

- https://funduinoshop.com/en/electronic-modules/valves-pumps/pumps/miniature-water-pump-3v-5v-100-200ma-1m-cable?srsltid=AfmBOooujcNcGX75lsYTr2hGbXYEtCPch-oHZywuD_OrPUEePF3Vcm4r
    - 3-5 V, 100-200 mA

- Wiring
    - https://www.youtube.com/watch?v=To3DKP99-1U

- Relay
    - https://www.amazon.ca/Huayao-Channel-Module-Shield-Arduino/dp/B07DN8WTKX
    - KY-019 5V Relay Module - ArduinoModulesInfo

ESP-32

- Enter deep sleep when no one is using?
- https://randomnerdtutorials.com/getting-started-with-esp32/

Other

- Glarks 120Pcs 2.54mm Straight Single Row PCB Board Female Pin Header Socket Connector Strip Assortment Kit for Arduino Prototype Shield(Single Row) : Amazon.ca: Tools & Home Improvement